



For the use of a Registered Medical Practitioner or a Hospital or a Laboratory Only.

Aerocort HFA Inhaler

Composition

Each actuation delivers
Salbutamol sulphate BP equivalent to
Salbutamol BP100 µg
Beclomethasone dipropionate BP 50 µg
Propellant HFA-134a

Description

AEROCORT HFA INHALER is a white homogeneous suspension aerosol for inhalation, in propellant HFA-134a, supplied in a pressurised container. On visual examination there should be no sign of physical damage, or leakage. This is CFC free metered dose Inhaler preparation.

Pharmacology

Pharmacodynamics

Salbutamol:

Salbutamol is a selective beta-2 adrenoceptor agonist. At therapeutic doses it acts on the beta-2 adrenoceptors of bronchial muscle.

Beclomethasone dipropionate:

Beclomethasone dipropionate given by inhalation has a potent glucocorticoid anti-inflammatory action within the lungs.

Pharmacokinetics

There is no evidence in human subjects that the administration of salbutamol and beclomethasone dipropionate together by the inhaled route in any way affects the pharmacokinetics of either component. Data in experimental animals indicate that inhaled Salbutamol does not affect the pharmacokinetics of simultaneously inhaled beclomethasone dipropionate and vice versa. For pharmacokinetic purposes therefore each component can be considered separately.

Salbutamol:

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally and partly by metabolism to the inactive 4'-O-sulphate (phenolic sulphate) which is also excreted primarily in the urine. The faeces are a minor route of excretion. The majority of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%. After administration by the inhaled route, between 10 and 20% of the dose reaches the lower airways. The remainder is retained in the delivery system or is deposited in the oropharynx from where it is swallowed. The fraction deposited in the airways is absorbed into the pulmonary tissues and circulation but it is not metabolised by the lung. On reaching the systemic circulation it becomes accessible to hepatic metabolism and is excreted, primarily in the urine, as unchanged drug and as phenolic sulphate.

The swallowed portion of an inhaled dose is absorbed from the gastrointestinal tract and undergoes considerable first pass metabolism to the phenolic sulphate. Both unchanged drug and conjugate are excreted primarily in the urine.

Beclomethasone dipropionate:

Beclomethasone 17, 21-dipropionate (BDP) administered intravenously is cleared rapidly with a half-life of approximately 30 minutes. Beclomethasone 17-monopropionate (BMP) appears rapidly in the plasma after intravenous BDP and is itself cleared with a half-life again of about 30 minutes. BDP is bound to plasma proteins to the extent of 87%. Up to 14% of an intravenous dose of BDP is excreted in the urine in 96 hours, mainly as polar metabolites, a proportion of which are conjugated. Up to 64% of the dose is excreted in faeces in his time, again primarily as free and conjugated metabolites.

After an inhaled dose, the majority of the dose will be deposited in the oropharynx or on the delivery device, with only a fraction reaching the lungs and available for absorption from this site. That portion deposited in the mouth or upper airways will ultimately be swallowed.

There is rapid metabolic inactivation of the majority of the swallowed portion of BDP during its first passage through the liver. An oral dose (4mg) of tritium labeled BDP was absorbed slowly with peak levels of radioactivity equivalent to 20ng drug/ml plasma being reached 5 hours after dosing. Excretion was mainly in the faeces (35-76% of the dose in 96 hours) and primarily as polar metabolites, although the presence of BDP and BMP in the faeces suggested incomplete absorption of the dose. Up to 14% of the dose was excreted as polar metabolites in urine.

The lung tissue rapidly hydrolyses BDP to BMP which in turn is hydrolyzed more slowly to beclomethasone. The liver also metabolizes BDP to BMP and further converts it to polar metabolites.

Indications

Aerocort Inhaler is suitable for those patients who require regular doses of both salbutamol and beclomethasone for treatment of reversible airways obstruction. Aerocort Inhaler is not intended for use as first-line treatment but is for use once the need for inhaled corticosteroid therapy has been established.

Dosage & Administration

Patients should be made aware that **Aerocort** Inhaler should be used regularly for optimum benefit. However, patients should be regularly reassessed so that their continuing need for corticosteroid therapy can also be reviewed.

Salbutamol is a selective beta2 agonist providing short acting (4-6 hours) bronchodilation with fast onset (within 5 minutes) in reversible airways obstruction.

Beclomethasone dipropionate provides effective anti-inflammatory action in the lungs, with a lower incidence and severity of adverse effects than those observed when corticosteroids are administered systemically. It also offers preventive treatment in asthma.

The dose should be titrated to the lowest dose at which effective control of asthma is maintained.

Adults:

2 inhalations (200 mcg salbutamol and 100 mcg beclomethasone dipropionate), three or four times a day.

The maximum dose is two inhalations four times a day. The usual starting dose is 2 inhalations three times a day.

Children:

1 or 2 inhalations (100 mcg to 200 mcg salbutamol and 50 mcg to 100 mcg beclomethasone dipropionate); two, three or four times a day.

The maximum dose is two inhalations four times a day.

The usual starting dose is two inhalations twice daily.

Elderly:

There is no need to adjust the dose in elderly patients.

Route of Administration:

Respiratory (Inhalation)

Contraindications

Although intravenous salbutamol and occasionally salbutamol tablets are used in the management of premature labour uncomplicated by conditions such as placenta praevia, anti-partum hemorrhage or toxemia of pregnancy, Aerocort Inhaler is not appropriate for managing premature labour. Aerocort Inhaler is contraindicated in patients with a history of hypersensitivity to any of its components.

Pregnancy and Lactation

Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus. As with the majority of drugs, there is little published evidence of the safety of salbutamol in the early stages of human pregnancy, but in animal studies there was evidence of some harmful effects on the fetus at very high dose levels.

There is inadequate evidence of safety of beclomethasone dipropionate in human pregnancy. Administration of corticosteroids to pregnant animals can cause abnormalities of foetal development including cleft palate and intra-uterine growth retardation. There may therefore be very small risks of such effects on the human foetus. It should be noted however, that the foetal changes in animals occur after relatively high systemic exposure. Because beclomethasone dipropionate is delivered directly to the lungs by the inhaled route it avoids the high level of exposure that occurs when corticosteroids are given by systemic routes.

The use of Aerocort Inhaler in pregnancy requires that the possible benefit of the drugs be weighed against the possible hazards.

Lactation:

No specific studies examining the transference of beclomethasone dipropionate into the milk of lactating animals have been performed. It is reasonable to assume that beclomethasone dipropionate is secreted in milk, but at the dosages used for direct inhalation there is low potential for significant levels in breast milk. As salbutamol is probably also secreted in breast milk the use of Aerocort Inhaler in nursing mothers should be restricted to situations where it is felt that the expected benefit to the mother is likely to outweigh any potential risk to the neonate.

Side Effects

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged

periods. These may include adrenal suppression, growth retardation in children and adolescents, increase in bone mineral density, cataract and glaucoma. As with other inhalation therapy, paradoxical bronchospasm may occur with an immediate increase in wheezing after dosing. This should be treated immediately with a fast-acting inhaled bronchodilator. Aerocort Inhaler should be discontinued immediately, the patient assessed, and, if necessary, alter-native therapy instituted.

Hoarseness, mouth and throat irritation may occur. It may be helpful to rinse the mouth with water immediately after inhalation.

Hypersensitivity reactions, including angioedema, bronch-ospasm, hypotension and collapse have been reported very rarely. Rashes, urticaria, pruritus and erythema and oedema of the eyes, face, lips and throat, have also been reported.

Salbutamol:

Salbutamol may cause a fine tremor of skeletal muscle; usually the hands are most obviously affected. This effect is dose related and is common to all β -adrenergic stimulants. Headaches have occasionally been reported. Tachycardia, with or without peripheral vasodilatation, may rarely occur. In common with other β_2 -agonists, cardiac arrhythmias (including atrial fibrillation, supraventricular tachycardia and extrasystoles) have been reported in association with the use of salbutamol, usually in susceptible patients.

There have been very rare reports of muscle cramps. Potentially serious hypokalaemia may result from β_2 -agonist therapy.

As with other β_2 -agonists, hyperactivity in children has been reported rarely.

Beclomethasone dipropionate:

Candidiasis of the mouth and throat (thrush) occurs in some patients inhaling beclomethasone dipropionate, the incidence increases with doses greater than 400 mcg beclomethasone dipropionate per day. Patients with high blood levels of *Candida precipitins*, indicating a previous infection, are most likely to develop this complication. Such patients may find it helpful to rinse their mouth with water after using the inhaler. The condition usually responds to topical anti-fungal therapy without discontinuing treatment with Aerocort Inhaler.

Warnings & Precautions

Patients should be instructed in the proper use of the inhaler to ensure that the drug reaches the target area within the lungs. They should also be made aware that Aerocort Inhaler must be used regularly, every day, for optimum benefit. Patients should be regularly reassessed so that their continuing need for corticosteroid therapy can also be reviewed.

Aerocort Inhaler is not for use in acute attacks but for routine long-term management, so patients will require a fast and a short-acting inhaled bronchodilator to relieve acute symptoms. However, should the effect of the additional inhaled bronchodilator or the relief provided by the Aerocort Inhaler last for less than three hours, patients should be advised that this may indicate that their patients

should be advised that this may indicate that their condition is worsening and to seek medical advice, in case treatment with inhaled corticosteroid needs to be increased, or treatment with systemic corticosteroid needs to be started or increased. As there may be adverse effects associated with excessive dosing, the dosage of frequency of administration should only be increased on medical advice.

Severe asthma requires regular medical assessment, including lung function testing, as patients are at risk of severe attacks and even death. Increasing use of short-acting inhaled β_2 -agonists to relieve symptoms indicates deterioration of asthma control. If patients find that short acting relief bronchodilator treatment becomes less effective or they need more inhalations than usual, medical attention must be sought.

For those patients who are steroid-dependent it is advisable to commence therapy with beclomethasone dipropionate alone.

Patients who have been weaned in the previous few months from long-term systemic corticosteroids need special consideration until the hypothalamic pituitary adrenal (HPA) system has recovered sufficiently to enable the patient to cope with emergencies such as trauma, surgery or infections. Such patients should carry a steroid warning card indicating that they may need supplementary systemic steroid during periods of stress, until their adrenocortical function has become normal. These patients should also be given a supply of oral steroid to use in emergency when their airways obstruction worsens.

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include adrenal suppression, growth retardation in children and adolescents, decreased in bone mineral density, cataract and glaucoma. It is important therefore that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control of asthma is maintained.

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be reviewed with the aim of reducing the dose of inhaled corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. In addition, consideration should be given to referring the patient to a pediatric respiratory specialist.

Prolonged treatment with high doses of inhaled corticosteroids, particularly higher than the recommended doses, may result in clinically significant adrenal suppression. Additional systemic corticosteroid cover should be considered during periods of stress or elective surgery.

As with all inhaled corticosteroids, special care is necessary in patients with active or quiescent pulmonary tuberculosis.

Salbutamol should be administered with caution to patients suffering from thyrotoxicosis.

Salbutamol and non-selective β -blocking drugs, such as propranolol, should not normally be prescribed together.

Potentially serious hypokalaemia may result from β 2-agonist therapy, mainly from parenteral and nebulised administration. Particular caution is advised in acute severe asthma, as this effect may be potentiated by hypoxia and by concomitant treatment with xanthine derivatives, steroids and diuresis. Serum potassium levels should be monitored in such situations.

Treatment should not be stopped abruptly.

Drug Interactions

Salbutamol and non-selective beta-blocking drugs, such as propranolol, should not usually be prescribed together. Salbutamol products are not contraindicated in patients under treatment with monoamine oxidase inhibitors (MAOIs). There are no proven drug interactions with beclomethasone dipropionate.

Overdosage

Salbutamol

The preferred antidote for overdosage with salbutamol is a cardioselective, beta-blocking agent, but beta-blocking drugs should be used with caution in patients with a history of bronchospasm. Hypokalaemia may occur following overdose with salbutamol. Serum potassium levels should be monitored.

Beclomethasone dipropionate

Acute:

Inhalation of the drug in doses in excess of those recommended may lead to temporary suppression of adrenal function. This does not require emergency action. In these patients treatment should be continued at a dose sufficient to control asthma; adrenal function recovers in a few days and can be verified by measuring plasma cortisol.

Chronic:

Use of inhaled beclomethasone dipropionate in daily doses in excess of 1,500 mcg over prolonged periods may lead to adrenal suppression. Monitoring of adrenal reserve may be indicated. Treatment should be continued at a dose sufficient to control asthma.

Storage

Store below 30°C. Avoid storage in direct sunlight or heat. Protect from frost.

Shelf life

24 months

Packing/Pack Size

200 metered dose

Registration holder in Malaysia

CIPLA MALAYSIA SDN BHD
SUITE 1101,
AMCORP TOWER
AMCORP TRADE CENTRE
18 PERSIARAN BARAT,
46050 PETALING JAYA,
SELANGOR, MALAYSIA

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Date of Revision:

18 June 2025

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